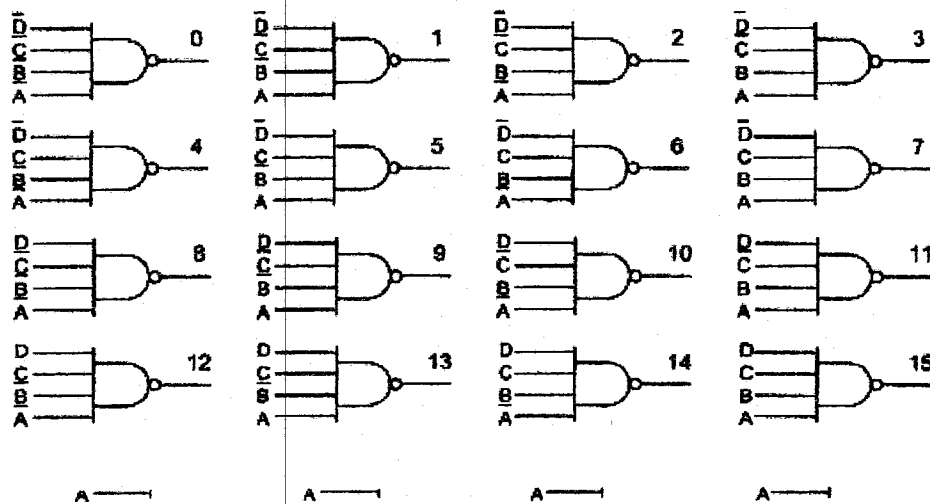
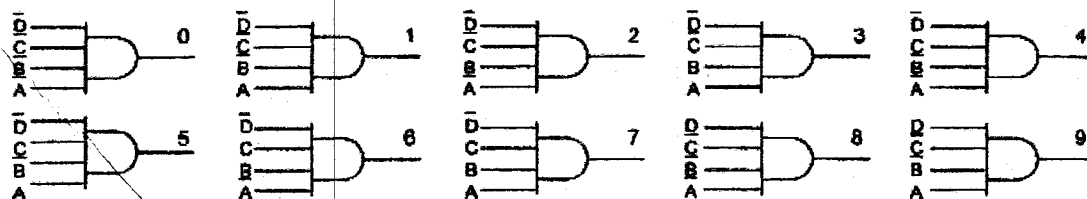


CHAPTER # HW

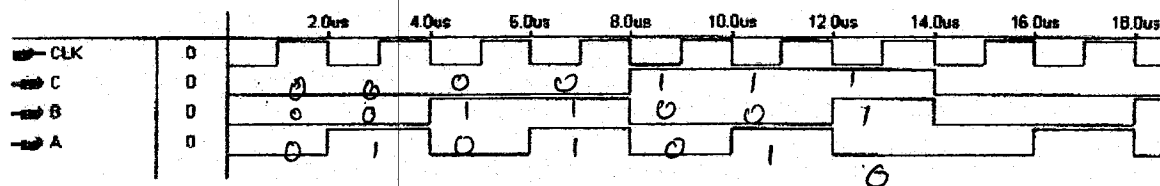
7.35



7.36



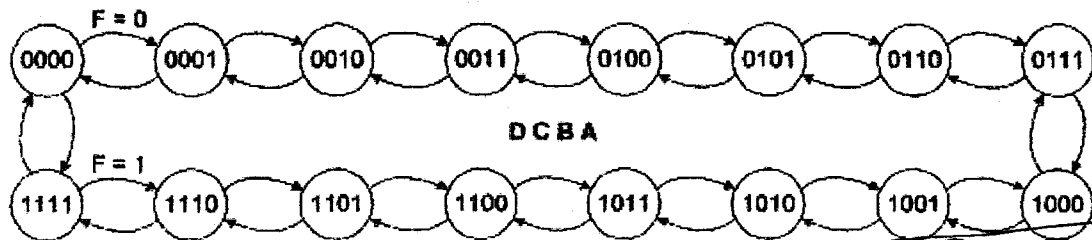
7.39 MOD-7 counter.



D is always 0. When it ~~tries~~ to ~~turn~~ + ~~again~~ At count 0111, the flip flops are all reset.

7.41

MOD-16 up/down counter.

Let $F=1$

$$J_A = K_A = 1$$

$$J_B = K_B = 1 \oplus A$$

$$J_C = K_C = (1 \oplus B)(1 \oplus A)$$

$$J_D = K_D = (1 \oplus B)(1 \oplus C)(1 \oplus A)$$

If $F=0$

$$J_A = K_A = 1$$

$$J_B = K_B = 0 \oplus A = A$$

$$J_C = K_C = AB$$

$$J_D = K_D = ABC$$

Means counter is
MOD 16 UP counter

For $F=1$

Present State	J_D	K_D	J_C	K_C	J_B	K_B	J_A	K_A	Next State
D C B A									D C B A
0 0 0 0	1	1	1	1	1	1	1	1	1 1 1 1
0 0 0 1									1 1 1 0
0 0 1 1	0	0	0	0	0	0	1	1	1 1 0 1
0 1 1 1	0	0	0	0	1	1	1	1	1 1 0 0
0 1 1 0	0	0	0	0	0	0	1	1	1 1 0 0
0 1 0 0	0	0	1	1	1	1	1	1	1 0 1 1
0 1 0 1	0	0	0	0	0	0	1	1	1 0 1 0
0 1 0 0									

As we continue it will work out to
be a down counter

7.43

(a) $J_A = B\bar{C}$, $K_A = 1$, $J_B = CA + \bar{C}\bar{A}$, $K_B = 1$, $J_C = B\bar{A}$, $K_C = B + \bar{A}$

~~(b) $J_A = AC$, $K_A = 1$, $J_B = KB = 1$, $J_C = KC = 1$~~

	J	K
$0 \rightarrow 0$	0	X
$0 \rightarrow 1$	1	X
$1 \rightarrow 0$	X	1
$1 \rightarrow 1$	X	0

	A	0	1
CB	00	0	0
	01	1	0
	11	X	X
	10	X	X

$$J_C = B\bar{A}$$

Present State

Next State

$CB A$	J_C	K_C	J_B	K_B	J_A	K_A	C	B	A
000	0	X	1	X	0	X	0	1	0
010	1	X	X	1	1	X	1	0	1
101	X	0	1	X	X	1	1	1	0
110	X	1	X	1	0	X	0	0	0
001	0	X	0	X	X	1	0	0	0
011	0	X	X	1	X	1	0	0	0
100	X	1	0	X	0	X	0	0	0
111	X	1	X	1	X	1	0	0	0

Use K-Map & Simplify to get answers in box. Use K-Map

